1. The energy density associated with electric field $\vec{E}$ and magnetic field $\vec{B}$ of an electromagnetic wave in free space is given by (ε_0- permittivity of free space, μ_0- permeability of free space)

[2023 (06 Apr Shift 2)]

(1) $U_E=\frac{E^2}{2\varepsilon_0}$, $U_B=\frac{B^2}{2\mu_0}$

(2) $U_E=\frac{\varepsilon_0 E^2}{2}$, $U_B=\frac{B^2}{2\mu_0}$

(3) $U_E=\frac{\varepsilon_0 E^2}{2}$, $U_B=\frac{\mu_0 B^2}{2}$

(4) $U_E=\frac{E^2}{2\varepsilon_0}$, $U_B=\frac{\mu_0 B^2}{2}$

2. The waves emitted when a metal target is bombarded with high energy electrons are

[2023 (08 Apr Shift 2)]

(1) Microwaves

(2) Infrared rays

(3) X-rays

(4) Radio Waves

3. The amplitude of magnetic field in an electromagnetic wave propagating along y-axis is 6.0×10^{-7} T. The maximum value of electric field in the electromagnetic wave is

[2023 (10 Apr Shift 2)]

(1) 2×10^{15} V m^{-1}

(2) 180 V m^{-1}

(3) 6.0×10^{-7} V m^{-1}

(4) 5×10^{14} V m^{-1}

4. A plane electromagnetic wave of frequency 20 MHz propagates in free space along x-direction. At a particular space and time $\vec{E}=6.6\hat{j}$ V m^{-1}. What is $\vec{B}$ at this point?

[2023 (11 Apr Shift 2)]

(1) $2.2\times 10^{-8}\hat{k}$T

(2) $-2.2\times 10^{-8}\hat{i}$T

(3) $-2.2\times 10^{-8}\hat{k}$T

(4) $2.2\times 10^{-8}\hat{i}$T

5. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : EM waves used for optical communication have longer wavelengths than that of microwave, employed in Radar technology.

Reason R : Infrared EM waves are more energetic than microwaves, (used in Radar)

In the light of given statements, choose the correct answer from the options given below.

[2023 (12 Apr Shift 1)]

(1) Both A and R are true but R is NOT the correct explanation of A

(2) A is false but R is true

(3) A is true but R is false

(4) Both A and R are true and R is the correct explanation of A

6. Which of the following Maxwell's equation is valid for time varying conditions but not valid for static conditions:

[2023 (13 Apr Shift 1)]

(1) $\oint\vec{B}\cdot\vec{dl}=\mu_0 I$

(2) $\oint\vec{E}\cdot\vec{dl}=0$

(3) $\oint\vec{D}\cdot\vec{dA}=Q$

(4) $\oint\vec{E}\cdot d\vec{l}=-\frac{\partial\phi_B}{\partial t}$

7. Given below are two statements:

Statement I : Out of microwaves, infrared rays and ultraviolet rays, ultraviolet rays are the most effective for the emission of electrons from a metallic surface

Statement II : Above the threshold frequency, the maximum kinetic energy of photoelectrons is inversely proportional to the frequency of the incident light

In the light of above statements, choose the correct answer from the options given below

[2023 (13 Apr Shift 2)]

(1) Statement I is false but Statement II is true

(2) Statement I is true but Statement II is false

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

8. In an electromagnetic wave, at an instant and at a particular position, the electric field is along the negative z-axis and magnetic field is along the positive x-axis. Then the direction of propagation of electromagnetic wave is:

[2023 (13 Apr Shift 2)]

(1) positive z-axis

(2) positive y-axis

(3) at 45° angle from positive y-axis

(4) negative y-axis

9. Match List-I with List II of Electromagnetic waves with corresponding wavelength range:

	List I		List II
(A)	Microwave	(I)	400 nm to 1 nm
(B)	Ultraviolet	(II)	1 nm to 10^{-3} nm
(C)	X-Ray	(III)	1 mm to 700 nm
(D)	Infra-red	(IV)	0.1 m to 1 mm

Choose the correct answer from the options given below:

[2023 (15 Apr Shift 1)]

(1) (A)–(IV), (B)–(I), (C)–(II), (D)–(III)

(2) (A)–(IV), (B)–(I), (C)–(III), (D)–(II)

(3) (A)–(IV), (B)–(II), (C)–(I), (D)–(III)

(4) (A)–(I), (B)–(IV), (C)–(II), (D)–(III)

ANSWER KEYS

1. (2)	**2.** (3)	**3.** (2)	**4.** (1)	**5.** (2)	**6.** (4)	**7.** (2)	**8.** (4)
9. (1)							

1. (2)

When an electromagnetic wave propagates from the source, it transfers energy to the objects in its path. Electric and magnetic fields serve as energy reservoirs for electromagnetic waves. An electromagnetic wave contains the same amount of energy as are contained in the electric and magnetic fields together. In such situation, the energy density of the electromagnetic wave is equal to the sum of the energies of the electric and magnetic fields.

The average electric and magnetic energy densities are given by

$U_E = \frac{1}{2}\varepsilon_0 E^2$

$U_B = \frac{1}{2}\frac{B^2}{\mu_0}$

2. (3)

The bombarding electrons can eject electrons from the inner shells of the atoms of the metal target. Those vacancies will be quickly filled by electrons dropping from higher levels, emitting x-rays with sharply defined frequencies associated with the difference between the atomic energy levels of the target atoms. Hence, X-rays are emitted when a metal target is bombarded with high energy electrons.

3. (2)

The relation between the electric field and magnetic field is given by

$E_0 = cB_0$.

The given data is

$B_0 = 6 \times 10^{-7}$ T

$c = 3 \times 10^8$ m s^{-1}

Thus, the maximum value of the electric field is

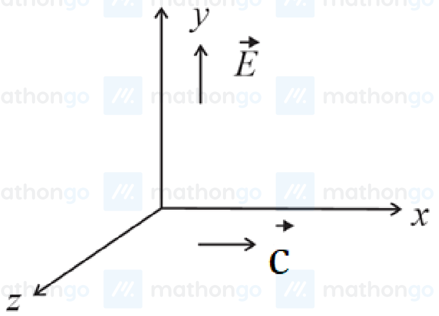
$E_0 = (3 \times 10^8) \times (6 \times 10^{-7}) = 180$ V m^{-1}.

4. (1)

It is given that

$\vec{E} = 6.6\hat{j}\left(\text{V m}^{-1}\right)$.

The relation between the electric, magnetic field and speed of light is $\vec{E} \times \vec{B} = \vec{c}$.



Clearly, $\vec{B} = B_0\left(\hat{k}\right)$

Using the relation,

$cB_0 = E$

$\Rightarrow B_0 = \left(\frac{E}{c}\right) = \frac{6.6}{3 \times 10^8} = 2.2 \times 10^{-8}$ T

5. (2)

An optical telecommunication is communication at a distance using light to carry information. It can be performed visually or by using electronic devices.
The wavelength of the microwaves are greater than that of the common optical light waves.
Hence, the Assertion is not correct. But, because of the small wavelength, the electromagnetic waves are more energetic than the microwaves.
Hence, the reason is true.

6. (4)

Maxwell's four equations describe the electric and magnetic fields arising from distributions of electric charges and currents, and how those fields change in time.
The equations are as follows:

$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\varepsilon_0} \quad \ldots(1)$

$\oint \vec{B} \cdot d\vec{A} = 0 \quad \ldots(2)$

$\oint \vec{E} \cdot d\vec{l} = -\frac{\partial \phi_B}{\partial t} \quad \ldots(3)$

$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad \ldots(4)$

From the above set of equations, it is clear that only the third equation is valid for time varying field.
Hence, this is the correct option.

7. (2)

Out of the three types of radiation, UV rays would be having highest frequency. The maximum kinetic energy is

$K_{\max} = hf - \phi$

Clearly, the maximum kinetic energy is dependant on the frequency of the incident radiation and the work function of the metal.
From the given spectrums UV rays have the highest energy. Hence, this option is the correct one.

8. (4)

Direction of propagation of the electromagnetic wave is along $\vec{E} \times \vec{B}$

Thus,

$\vec{n} = E\left(-\hat{k}\right) \times B\left(\hat{i}\right)$

Therefore,

$\hat{n} = -\hat{k} \times \hat{i} = -\hat{j}$

9. (1)

The electromagnetic spectrum is the range of all types of EM radiation. Radiation is energy that travels and spreads out with different energies. In the daily life, two commonly known EM waves are the light wave and the radio wave.
The other types of EM radiation that make up the electromagnetic spectrum are microwaves, infrared light, ultraviolet light, X-rays and gamma-rays.
The wavelengths of different parts of the electromagnetic spectrum are related by the following relation:

$\lambda_{\mathrm{X}} < \lambda_{\mathrm{UV}} < \lambda_{\mathrm{IR}} < \lambda_{\mathrm{MW}}$